



Research Paper

The relevance of two-phase flow in the thermo-hydro-mechanical evolution of clay formations exposed to high temperatures by heat-emitting waste

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ABSTRACT

We compare two-phase flow and Richards flow implementations in OpenGeoSys-6 to model the thermo-hydro-mechanical evolution of heat-emitting waste in clay stone formations. Our quasi-1D example is based on the material sequence and domain properties observed in the FE experiment at the Mt. Terri underground research lab in Switzerland. We examine the validity of the Richards assumption by comparing a thermo-hydro-mechanically (THM) coupled Richards model against two THM-coupled two-phase flow-based models, one where the gas pressure is constrained to atmospheric pressure, and one unconstrained model. The model comparison was conducted with saturation-dependent permeability models at temperatures up to $\approx 200^\circ\text{C}$. Additionally, we consider the impact of two different vapor diffusion models, a gas pressure-independent empirical relationship versus the original De Vries model, which becomes relevant if gas pressure buildup is significant.

Our results show excellent agreement between the two models for maximum temperatures around 100°C . Even at higher temperatures, above 150°C , we observe good agreement, which improves significantly with increasing distance from the heater. Even for the highest heat power where both approaches differ significantly in the high-temperature regions, acceptable agreement can be reached outside those regions, i.e. a couple of tens of centimeters away from the heater, but still in the bentonite barrier domain. This work builds confidence in the use of Richards-based approaches for modeling of the THM processes in nuclear waste repository, and contributes to a knowledge-driven model selection in the context of safety-relevant radioactive waste management.

1. Introduction

Recently, several countries have submitted their site proposals for the construction of deep geological repositories (DGRs) for the deposition of high-level radioactive waste. In this context, reliability and efficiency of computational models capable of simulating THMC processes in the near- and far-fields, as well as for entire repository systems, have a high priority in related safety and performance assessments. In addition to activities at the national level, international projects such as DECOVALEX [1–3], EURAD [4] or the FE Task Force [5] exist to ensure the exchange of knowledge, including the validation and verification of the numerical methods at hand. Some countries, such as Germany, consider different host rock types

as potential DGRs, which makes comparability of those options also a challenging modeling task [6–9]. In addition, additional challenges to software development and professional engineering, such as the FAIR¹ [10,11] principles, are becoming more and more important for scientific software projects, particularly in safety-relevant applications such as radioactive waste deposition [12].

The modeling of entire bentonite-backfilled repositories for high-level radioactive waste using multiphase thermo-hydro-mechanical models is still computationally demanding. Therefore, several simplifications such as condensed elasticity models [13,14], surrogate modeling [15–17], or the Richards assumption [18] which has been widely used to study unsaturated media [19–22], have been successfully adopted to speed up calculations. In this publication, we will focus

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¹ Findable, accessible, interoperable, and reproducible.

on the latter approach that is the most widely used. Despite its broad usage, it is not a priori clear whether the underlying assumptions are justified in many cases and whether simulations yield correct results under any given circumstance. Although there are a few studies that compare the multiphase and Richards approaches [23–25] and show good agreement if the intrinsic permeability for the gaseous phase is sufficiently high, a detailed study exploring their limits, for example, at high temperatures (as explored as part of the HotBent project [26]) or once advanced bentonite resaturation has already taken place, is still outstanding. In particular, temperature ranges of up to 200 °C have gained interest in the context of bentonite barriers [26,27]. We need to mention that this comparison is also motivated by the fact that the theoretical description of multiphase and Richards approaches used throughout this publication do not completely coincide in the high-gas permeability limit, as shown by Pitz et al. [24], since both were derived separately. In this context, it is also important to consider the saturation dependence of the permeability [23,28] as it might lead to a local (in time and space) violation of the Richards assumption.

In this publication, we discuss the applicability of the Richards assumption by comparing the Thermo-Richards-Mechanics (TRM) model against a newly implemented non-isothermal two-phase (TH²M) model [29]. Different design temperatures are discussed to reduce repository area requirements and mined volumes [27]. Therefore, models that were successful in low-temperature contexts need to be qualified in a high-temperature context as well. A first step is a comparative study of current model generations. Both models are part of the OpenGeoSys-6 software platform [30], a computer code to model thermo-hydro-mechanical-chemical (THMC) processes in porous and fractured media. Although there are several other codes available for simulating multiphase flow (e.g., Moose [31], CODE_BRIGHT [32], TOUGH-FLAC [33]), OpenGeoSys (OGS) is open source and both models are assembled in OGS as a monolithic scheme, providing more accurate and stable results than their staggered coupled counterparts. Indeed, such benefits are not without cost, as they are considered less efficient under certain conditions. However, we chose OpenGeoSys because it is an efficient open-source code that has implemented both approaches and is also very flexible in terms of dealing with custom-tailored material models.

To our knowledge, there are no comparative studies for both approaches, with a focus on their limits including long simulation times (high bentonite saturations), high temperatures, and the impact of two commonly used vapor diffusion models at hand. We think that our study can build confidence in the numerically more efficient Richards approaches, while points out under which circumstances and for which quantities of interest multiphase approaches might be preferred in an area that is highly relevant to safety and reliability.

Our benchmark case is based on the FE-experiment conducted in the Mont Terri underground research lab in Switzerland. The original experiment “*simulates, as realistically as possible, the construction, waste emplacement, backfilling and early post-closure evolution of a spent fuel/vitrified high-level waste disposal tunnel according to the Swiss repository concept. The main aim of this multiple heater test is the investigation of repository-induced thermo-hydro-mechanical (THM) coupled effects on the host rock at this scale and the validation of existing coupled THM models. For this, several hundred sensors were installed in the rock, the tunnel lining, the bentonite buffer, the heaters, and the plug.*” [34]

Both theoretical approaches, multiphase and Richards’, were successfully used to model the thermo-hydro-mechanical evolution of the bentonite barrier and the host rock in the FE-experiment [5,17,35,36]. Although good agreement between the experimental and numerical results for the temperature response was usually found, there is some room for improvement with respect to an accurate pressure prediction. This is not necessarily surprising, as the original experiment in its entirety has many features that render its modeling complex, such as combined construction and operating phases, multiple materials

exhibiting different properties at different stages, anisotropy, and geometric complexity, to name but a few. Their correct representation is a challenging and computationally demanding task. Under these conditions, it is all the more important to compare different models that represent the experiment. Therefore, we restrict our investigation to a simple quasi-1D case with multiple material domain sections, as we are focusing our comparison solely on the non-isothermal two-phase/Richards flow in THM models. This simplification to 1D is crucial to reduce computational complexity and to isolate and better understand fundamental processes without the added intricacies of a full 3D model. Although both of our models include a mechanical coupling in the form of a linear elasticity model, we omit its detailed analysis, as it is believed to be in complete agreement. The swelling effects that also become relevant with the changing water content and depend on multiple factors such as bentonite composition and water salinity [37] are also neglected for this study.

2. Governing equations

This publication compares the results of two recent implementations in the continuum porous media modeling framework OpenGeoSys 6 [38]: a non-isothermal two-phase two-component flow with the mechanics model (TH²M) and a non-isothermal Richards mechanics (TRM) model. Both models describe the evolution of temperature, pressure, and displacement of fluid and solid phases in an unsaturated porous medium. The models are based on the assumption that the porous medium is continuous and deformable, with pores filled by both liquid and gas. In both models, the hydro-mechanical coupling follows from Biot’s consolidation theory [39] and is handled in a similar manner. The TH²M model considers two phases, a liquid and a gaseous phase with two components, water and non-aqueous, each. This allows for vapor diffusion and advection and phase transitions between the liquid and the gaseous phase to take place. The TRM model, on the other hand, is based on the Richards approximation that assumes that the mobility of the gaseous phase is high enough such that there is no relevant gas pressure build-up taking place; i.e., the gas pressure is assumed to be constant. In contrast to the initial assumption, the Richards equation was expanded to include the water vaporization process and the diffusive transport of water by permitting vapor pressure gradients and the resulting vapor diffusion in the gas phase [40].

Both implementations were verified and presented in earlier publications [24,29,41,42]. In the following chapter, we will briefly review the governing equations which are solved for the primary variables temperature T , capillary pressure p_{cap} , gas pressure p_{GR} , and solid displacement vector u_s .

2.1. Mass balance equations

2.1.1. TH²M mass balance

For the two-phase flow mass balance, we assume that the pores of a deformable porous solid $\alpha \in S$ are filled with fluids present in a liquid and a gaseous phase, $\alpha \in \{L, G\}$. The general formulation assumes the presence of up to two components in each phase: If phase transitions are excluded, no components are considered, and the governing equations simplify to a classical two-phase flow formulation. However, phase transitions could be considered for either a dissolving gas component or an evaporating water component, or both. The detailed derivation of this formulation can be found in [29].

In the general case, each fluid phase α is populated by two components ζ and the component-based mass balance reads as

$$0 = \rho_{\text{FR}}^{\zeta} (\alpha_B - \phi) \beta_{p,\text{SR}} \frac{dp_{\text{GR}}}{dt} - \rho_{\text{FR}}^{\zeta} (\alpha_B - \phi) \beta_{p,\text{SR}} S_L \frac{dp_{\text{cap}}}{dt} - \rho_{\text{FR}}^{\zeta} (\alpha_B - \phi) \text{tr}(\alpha_{T,\text{SR}}) \frac{dT}{dt} \\ + \phi \left[(1 - S_L) \frac{d\rho_{\text{GR}}^{\zeta}}{dt} + S_L \frac{d\rho_{\text{LR}}^{\zeta}}{dt} \right] + \left[\phi (\rho_{\text{LR}}^{\zeta} - \rho_{\text{GR}}^{\zeta}) - \rho_{\text{FR}}^{\zeta} p_{\text{cap}} (\alpha_B - \phi) \beta_{p,\text{SR}} \right] \frac{dS_L}{dt} \\ + \rho_{\text{FR}}^{\zeta} \alpha_B \text{div} \left(\frac{du_s}{dt} \right) + \text{div}(\mathbf{A}^{\zeta} + \mathbf{J}^{\zeta}), \quad (1)$$

with material time derivative $\frac{d\circ}{dt} = \frac{\partial\circ}{\partial t} + \mathbf{v}_S \cdot \text{div } \circ$ of the quantity $\circ$ with respect to the velocity of the solid phase $\mathbf{v}_S$.

The terms α_B and ϕ represent the Biot–Willis coefficient and porosity, respectively, accounting for the compressibility of the solid grains and the fraction of the volume occupied by voids. While α_B reflects the proportion of total stress carried by the pore fluid, ϕ quantifies the storage capacity of the porous medium.

The parameter $\beta_{p,SR}$ is the compressibility of the solid grain, while $\alpha_{T,SR}$ is a diagonal matrix containing the linear thermal expansion coefficients of the solid in all coordinate directions. Thus, $\text{tr } \alpha_{T,SR}$ provides the volumetric thermal expansion coefficient of the solid. The saturation of the liquid phase, S_L , represents the fraction of the pore volume occupied by the liquid phase.

The pressure of the liquid phase can be expressed in terms of the primary variables as the difference between gas pressure and capillary pressure,

$$p_{LR} = p_{GR} - p_{cap}, \quad (2)$$

and the effective partial density of the component ζ in both fluid phases is given by

$$\rho_{FR}^\zeta = (1 - S_L)\rho_{GR}^\zeta + S_L\rho_{LR}^\zeta, \quad (3)$$

where ρ_{GR}^ζ and ρ_{LR}^ζ are the partial densities of ζ in the gas and liquid phases.

Advective fluxes $\mathbf{A}^\zeta = \mathbf{A}_L^\zeta + \mathbf{A}_G^\zeta$ are given by

$$\mathbf{A}_\alpha^\zeta = \rho_{\alpha R}^\zeta \tilde{\mathbf{w}}_{\alpha S} = -\rho_{\alpha R}^\zeta \frac{k_{\alpha}^{\text{rel}} \mathbf{k}_S}{\mu_{\alpha R}} (\text{grad } p_{\alpha R} - \rho_{\alpha R} \mathbf{g}), \quad (4)$$

where $\tilde{\mathbf{w}}_{\alpha S}$ resembles the velocity of fluid α relative to the solid phase, as described by Darcy's law, whereas the diffusive–dispersive fluxes $\mathbf{J}^\zeta = \mathbf{J}_L^\zeta + \mathbf{J}_G^\zeta$ of component ζ read

$$\mathbf{J}_G^\zeta = -\rho_{GR} D_G^\zeta \text{grad } x_{m,G}^\zeta = -\rho_{GR} D_G^\zeta \left(\frac{\partial x_{m,G}^\zeta}{\partial p_{GR}} \text{grad } p_{GR} + \frac{\partial x_{m,G}^\zeta}{\partial p_{cap}} \text{grad } p_{cap} + \frac{\partial x_{m,G}^\zeta}{\partial T} \text{grad } T \right) \quad (5)$$

for the gaseous phase and

$$\begin{aligned} \mathbf{J}_L^\zeta &= -\rho_{LR} D_L^\zeta \text{grad } x_{m,L}^\zeta \\ &= -\rho_{LR} D_L^\zeta \left(\frac{\partial x_{m,L}^\zeta}{\partial p_{LR}} (\text{grad } p_{GR} - \text{grad } p_{cap}) + \frac{\partial x_{m,L}^\zeta}{\partial p_{GR}} \text{grad } p_{GR} + \frac{\partial x_{m,L}^\zeta}{\partial T} \text{grad } T \right) \end{aligned} \quad (6)$$

for the liquid phase, with $x_{m,\alpha}^\zeta$ as the molar fraction of component ζ in phase α .

In the general case, (1) is applied separately to each component, $\zeta = C$ and $\zeta = W$. This results in two equations, which are perfectly symmetric when considering phase transitions in the form of both dissolution/outgassing and evaporation/condensation.

Although this work compares two-phase flow simulations with Richards simulations, the focus is on vapor diffusion, which is relevant over short time scales. The Richards approach does not explicitly consider a gas phase, which simplifies the treatment of the liquid phase in this context.

Specifically, this absence reduces the need to consider advection and diffusion terms for a dissolved gas component, altering the symmetry of the equations and leading to distinct handling for each component. Diffusive transports of dissolved gases, relevant over longer time scales, require a two-component formulation and are not considered in this analysis.

2.1.2. TRM mass balance

The non-isothermal Richards flow with mechanics model [23,24] as implemented in OGS-6 is formulated in terms of a single mass balance equation for water with liquid pressure p_{LR} . To account for vapor-related effects like desaturation due to temperature gradients, an

additional vapor flux is introduced according to the formalism derived by Philip and DeVries [43]. The mass balance can then be written as:

$$\begin{aligned} 0 &= \rho_{LR} S_L^2 (\alpha_B - \phi) \beta_{p,SR} \frac{dp_{LR}}{dt} - \rho_{LR} S_L (\alpha_B - \phi) \text{tr } (\alpha_{T,SR}) \frac{dT}{dt} \\ &+ \phi \left((1 - S_L) \frac{d\rho_{GR}^W}{dt} + S_L \frac{d\rho_{LR}^W}{dt} \right) + (\rho_{LR} - \rho_{GR}^W) [\phi + p_{LR} S_L (\alpha_B - \phi) \beta_{p,SR}] \frac{dS_L}{dt} \\ &+ S_L \rho_{LR} \alpha_B \text{div } \left(\frac{d\mathbf{u}_S}{dt} \right) + \text{div } (\mathbf{A}_L^W + \mathbf{J}_G^W), \end{aligned} \quad (7)$$

with liquid phase density ρ_{LR} . The advective flux $\mathbf{A}_L^W$ agrees with the same contribution in the TH²M model. For TRM, we assume that $p_{cap} = -p_{LR}$. The liquid saturation S_L is calculated from the van Genuchten water retention relation. Although no separate gas phase is present in this model, we consider a contribution from the vapor flux $\mathbf{J}_G^W$, which is calculated from:

$$\mathbf{J}_G^W = -D_G^W \text{grad } \rho_{GR}^W = -D_G^W \left(\frac{\partial \rho_{GR}^W}{\partial p_{LR}} \text{grad } p_{LR} + \frac{\partial \rho_{GR}^W}{\partial T} \text{grad } T \right). \quad (8)$$

where the partial density of the water component in the gas phase, ρ_{GR}^W (vapor density), is derived from saturated vapor densities ρ_{vap}^W , as detailed by Eqs. (24) and (25) in Section 2.4.4.

2.2. Momentum balance equations

Given the effective stress tensor $\boldsymbol{\sigma}' = \mathbf{C} : (\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_{\text{inelastic}} - \boldsymbol{\epsilon}_{\text{thermal}})$, we can write the momentum balance equation following Biot. The thermal strain rate is given by $\frac{d}{dt} \boldsymbol{\epsilon}_{\text{thermal}} = \alpha_{T,SR} \frac{dT}{dt}$. For TH²M, we have

$$\mathbf{0} = \text{div } (\boldsymbol{\sigma}' - \alpha_B [\chi p_{LR} + (1 - \chi) p_{GR}] \mathbf{I}) + [(1 - \phi) \rho_{SR} + S_L \phi \rho_{LR} + (1 - S_L) \phi \rho_{GR}] \mathbf{g} \quad (9)$$

with Bishop coefficient χ and the solid phase density ρ_{SR} , while for TRM, it reads:

$$\mathbf{0} = \text{div } (\boldsymbol{\sigma}' - \alpha_B \chi p_{LR} \mathbf{I}) + [(1 - \phi) \rho_{SR} + S_L \phi \rho_{LR}] \mathbf{g}. \quad (10)$$

Both equations differ only regarding the consideration of pore gas pressure and pore gas density. The Bishop coefficient χ is typically chosen to be saturation-dependent. For the purposes of this paper, we use a power law

$$\chi = S_L^\lambda \quad (11)$$

with exponent $\lambda = 5$ for bentonite and shotcrete and $\lambda = 8$ for OPA, based on the rationale that finer-grained soils lead to higher values for λ [44].

2.3. Energy balance equations

2.3.1. TH²M enthalpy balance

The energy balance in the TH²M implementation is given by Grunwald et al. [29]:

$$\begin{aligned} 0 &= \underbrace{\frac{d}{dt} (\sum_a \rho_a u_a)}_{\text{storage}} + \underbrace{\text{div } (\sum_a h_a \mathbf{A}_a)}_{\text{transport by advection}} + \underbrace{\text{div } (\sum_a \sum_\zeta h_a^\zeta \mathbf{J}_a^\zeta)}_{\text{transport by diffusion}} + \underbrace{(\sum_a \rho_a h_a) \text{div } \left(\frac{d\mathbf{u}_S}{dt} \right)}_{\text{medium volume change}} - \underbrace{\sum_a \mathbf{g} \cdot \mathbf{A}_a}_{\text{gravitation work}} \\ &\quad - \underbrace{\text{div } (\lambda_{\text{eff}} \text{grad } T)}_{\text{heat conduction}}, \end{aligned} \quad (12)$$

where ρ_a is the apparent density of a phase, which, expressed for each phase, reads $\rho_L = \phi S_L \rho_{LR}$, $\rho_G = \phi(1 - S_L) \rho_{GR}$, and $\rho_S = (1 - \phi) \rho_{SR}$, respectively. u_a is the specific internal energy of each phase and $h_a = \frac{p_a}{\rho_a} + u_a$ is the specific enthalpy. From the definitions of u_a and h_a , one can observe that pressure work is included in the formulation. The advective mass flux of α represents the phase mixtures containing both components, and it is given by $\mathbf{A}_\alpha = \rho_{\alpha R} \tilde{\mathbf{w}}_{\alpha S}$ with the Darcy velocity

defined according to Eq. (4).

The effective medium thermal heat capacity, based on the liquid, gas, and solid phase properties, is averaged using the common mixing rule:

$$(\rho c_p)_{\text{eff}} = \sum_{\alpha} \rho_{\alpha} c_{p\alpha} \quad (13)$$

where the phase heat capacities are in turn calculated using a mixing rule according to the phase compositions $c_{p\alpha} = \sum_{\alpha} x_{m,\alpha}^{\zeta} c_{p\alpha}^{\zeta}$. The storage term in Eq. (12) can be expanded, showing that it reflects both enthalpy stored due to temperature changes and enthalpy stored due to phase pressure changes and phase transitions [24].

2.3.2. TRM energy balance

In TRM, the energy balance equation reads:

$$0 = (\rho c_p)_{\text{eff}} \frac{dT}{dt} + L_0 \frac{d\theta_{\text{vap}}}{dt} - \text{div}(\lambda_{\text{eff}} \text{grad } T) + \text{div} \left(\frac{L_0 \mathbf{J}_G^W}{\rho_{\text{LR}}} \right) + \text{grad } T \cdot (c_{pL} \mathbf{A}_L + c_{p,\text{vap}} \mathbf{J}_G^W), \quad (14)$$

with the volumetric vapor content defined by De Vries [45]:

$$\theta_{\text{vap}} = \phi(1 - S_L) \frac{\rho_{\text{GR}}^W}{\rho_{\text{LR}}}. \quad (15)$$

This formulation can be derived from the more general enthalpy balance Eq. (12) (see Pitz et al. [24]). The volumetric latent heat of water vaporization [46], given in J m^{-3} , is:

$$L_0 = \rho_{\text{LR}} (2.501 \times 10^6 \text{ J kg}^{-1} - 2369.2 \text{ J kg}^{-1} \text{ K}^{-1} (T - 273.15 \text{ K})). \quad (16)$$

This equation accounts for the temperature dependence of the latent heat, ensuring accurate modeling of energy conservation in the system.

2.4. Constitutive equations

2.4.1. Liquid density

The liquid density is assumed to follow an exponential relationship [47]:

$$\rho_{\text{LR}} = \rho_{\text{LR,ref}} \exp(\beta_{p,\text{LR}} (\max(0, p_{\text{LR}}) - p_{\text{LR,ref}}) - \beta_{T,\text{LR}} (T - T_{\text{ref}})) \quad (17)$$

where $\rho_{\text{LR,ref}}$ is the density at reference pressure $p_{\text{LR,ref}}$ and reference temperature T_{ref} . For the given example, we use $\rho_{\text{LR,ref}} = 1002.6 \text{ kg m}^{-3}$, $p_{\text{LR,ref}} = 0.1 \text{ MPa}$, and $T_{\text{ref}} = 273.15 \text{ K}$ with compressibility $\beta_{p,\text{LR}} = 4.5 \times 10^{-10} \text{ Pa}^{-1}$ and thermal expansion coefficient $\beta_{T,\text{LR}} = -2.0 \times 10^{-4} \text{ K}^{-1}$.

2.4.2. Liquid viscosity

Like the liquid density, we also account for the reduction in viscosity at high temperatures by fitting the IAPWS data to the formula of Andrade [48]:

$$\mu_{\text{LR}} = 2.1 \times 10^{-6} \text{ Pa s} \exp(1808.5 \text{ K}/T). \quad (18)$$

2.4.3. Water retention and relative permeability

The water retention behavior in this work is described according to the relation of van Genuchten [49] where liquid saturation S_L is a function of capillary pressure p_{cap} , which is described in terms of the effective saturation S_e with $S_L = S_e (S_{L,\text{max}} - S_{L,\text{res}}) + S_{L,\text{res}}$:

$$S_e = \left(1 + \left(\frac{p_{\text{cap}}}{p_e} \right)^n \right)^{\frac{1}{n-1}}, \quad (19)$$

where p_e represents the gas entry pressure and $n \in [0, 1)$ is an empirical shape parameter. The liquid relative permeability is defined according to van Genuchten/Mualem [50]:

$$k_L^{\text{rel}} = \sqrt{S_e} \left(1 - \left(1 - S_e^{n/(n-1)} \right)^{(n-1)/n} \right)^2. \quad (20)$$

The relative gas permeability k_G^{rel} , is described in terms of a generalized power law:

$$k_G^{\text{rel}} = 1.0 \times 10^6 (S_e)^4, \quad (21)$$

which was determined from fitting the permeability data presented in [28]. According to Olivella and Gens [28] the difference between liquid water permeability and gas permeability can be at least partially explained by the Knudsen or Klinkenberg [51] effects.

2.4.4. Vapor pressure/vapor density model

To obtain comparable results between the Richards and the two-phase flow models, we used the following expression for the water vapor pressure in the TH²M model:

$$p_{\text{GR}}^W = p_{\text{vap}}^W \exp \left(- \frac{p_{\text{cap}} M_W}{\rho_{\text{LR}} R T} \right), \quad (22)$$

where the saturated vapor pressure is given by

$$p_{\text{vap}}^W = \frac{RT}{M_W} \rho_{\text{vap}}^W. \quad (23)$$

Analogously, in the TRM model we set the water vapor density to

$$\rho_{\text{GR}}^W = \rho_{\text{vap}}^W \cdot \exp \left(\frac{p_{\text{LR}}}{\rho_{\text{LR}} R T} \right). \quad (24)$$

For both models, the saturated vapor density is given by an empirical relationship [52]:

$$\rho_{\text{vap}}^W = 10^{-3} \frac{\text{kg}}{\text{m}^3} \cdot \exp(19.819 - 4975.9 \text{ K}/T). \quad (25)$$

2.4.5. Vapor diffusion models

One significant factor for the resaturation behavior of bentonite is the vapor diffusion model, which is why we examine the impact of two widely used approaches: one proposed by DeVries [43] and an empirical gas pressure-independent relationship (see e.g., Birkholzer et al. [53] where it is used to describe vapor diffusion in FEBEX type bentonite with tortuosity $\tau = 0.8$). For the former case, the vapor diffusivity D_G^W is given by

$$D_G^W = \tau \phi (1 - S_L) 5.9 \times 10^{-6} \text{ m}^2 \text{ s}^{-1} \left(\frac{T/\text{K}}{p_G/\text{Pa}} \right)^{2.3} \mathbf{I}, \quad (26)$$

whereas the latter can be written in terms of temperature only:

$$D_G^W = \tau \phi (1 - S_L) 2.16 \times 10^{-5} \text{ m}^2 \text{ s}^{-1} \left(\frac{T}{273.15 \text{ K}} \right)^{1.8} \mathbf{I}. \quad (27)$$

Both models were used for the TRM and TH²M modeling. For TRM, p_G was set to atmospheric pressure $p_{\text{atm}} = 0.1 \text{ MPa}$.

3. Computational details

3.1. Benchmark definition

Comparing both schemes requires a robust benchmark; therefore, we consider a variation of the benchmark based on the FE-experiment conducted in the Mont Terri underground research lab in Switzerland. This simplified quasi-1D case focuses on non-isothermal radial two-phase/Richards flow and addresses important issues in radioactive waste repository design, such as saturation of geotechnical barriers and pore pressurization in the geological barrier. The following sections detail the computational setup and parameters used for these simulations.

The model domain measures $50 \text{ m} \times 0.02 \text{ m}$ in radial and axial direction of the emplacement drift, respectively. The structure and dimensions of the model were derived from 3D models featured in previous works (see, e.g., Buchwald et al. [17]), providing the outer radii of the heater, the granular bentonite material (GBM), the shotcrete (SC), and the excavation damaged zone (EDZ) as 0.525 m , 1.4 m , 1.6 m , and 1.6 m , respectively. A sketch of the model domain, along with

Table 1
Material properties for different material groups. Modified after Thatcher et al. [54] and Buchwald et al. [17].

Property	Symbol	Unit	GBM	Shotcrete	EDZ	Clay
Solid density	ρ_s	kg m ⁻³	2500.0	2706	2690	2690
Specific heat capacity	c_s	J kg ⁻¹ K ⁻¹	800	750	1086	995
Thermal expansivity	α_s	K ⁻¹	3.0e-6	1.5e-5	1.7e-5	1.7e-5
Young's modulus	E	Pa	18e6	2e10	6e9	6e9
Poisson's ratio	ν	1	0.35	0.15	0.3	0.3
Thermal conductivity	λ	W m ⁻¹ K ⁻¹	$1.2^{S_L} \cdot 0.35^{1-S_L}$	$2.0S_L + 1.0(1 - S_L)$	1.9	1.9
Intrinsic permeability	k	m ²	3.5e-20	1e-19	3.0e-20	3.0e-20
Porosity	ϕ	1	0.331	0.25	0.13	0.13
Biot-Willis coefficient	α_B	1	1.0	1.0	1	1
Saturation: Van Genuchten, residual liquid saturation	S_{rL}^r	1	0.01	0.01	0.3	0.3
Saturation: Van Genuchten, residual gas saturation	S_{rG}^r	1	1e-15	1e-15	1e-15	1e-15
Saturation: Van Genuchten, exponent	m	1	0.45	0.33	0.4	0.4
Saturation: Van Genuchten, entry pressure	p_b	Pa	18e6	1.0e6	9.e6	20.e6
Relative permeability: Van Genuchten, residual liquid saturation	S_{rL}^r	1	0.01	0.01	0.3	0.3
Relative permeability: Van Genuchten, residual gas saturation	S_{rG}^r	1	1e-15	1e-15	1e-15	1e-15
Relative permeability: Van Genuchten, exponent	m	1	0.45	0.33	0.4	0.4
Relative permeability: Van Genuchten, minimum relative permeability	$k_r^{\min}$	1	1e-9	1e-9	1e-9	1e-9
Liquid specific heat capacity	c_{LR}	J kg ⁻¹ K ⁻¹	4280.0	4280.0	4280.0	4280.0

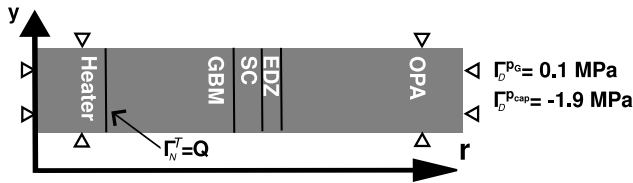


Fig. 1. Two-dimensional slice of the cylindrical model domain. The inner core consists of a heating element surrounded by GBM and SC as technical barriers in an Opalinus clay cavity.

the relevant boundary conditions, is shown in Fig. 1, while all remaining boundaries were set as natural zero flux (Neumann-zero) boundaries. All boundary conditions of the quasi-one-dimensional domain are aimed at reflecting the behavior of the three-dimensional model (e.g., Buchwald et al. [17] and Alcolea et al. [5]). All parameters for the different material groups are based on data provided to the modeling teams in previous modeling initiatives (DECOVALEX and the FE modeling Task Force) and are summarized in Table 1.

The initial conditions in Opalinus clay (OPA) are $T = 17^\circ\text{C}$, $p_{LR} = 2\text{ MPa}$ and $p_{GR} = 0.1\text{ MPa}$. For simplicity, we assumed $\sigma'_{ij} = 0.0\text{ MPa}$, as no excavation was modeled in this elastic setup, and rotational symmetry would conflict with the actual in-situ stresses. The initial capillary pressures in the GBM and shotcrete were chosen to obtain liquid saturation values of $S_{GBM} = 0.188$ and $S_{Shotcrete} = 0.383$, respectively. The corresponding temperatures and gas pressures were taken to be the same as in the OPA.

We considered three heat source powers: $Q_1 = 1350\text{ W}$, $Q_2 = 2700\text{ W}$, and $Q_3 = 3780\text{ W}$. In the original experiment, three canisters are stacked along a test section of about 23 m. Therefore, these heat powers were associated to the following heat densities: $q_1 = 53.385\text{ W m}^{-2}$, $q_2 = 106.770\text{ W m}^{-2}$ and $q_3 = 149.478\text{ W m}^{-2}$, which were applied as Neumann conditions at the heater circumference.

As shown in Table 2, several observation points are specified for variable output. However, in this study we focus mainly on GBM1 and GBM2 due to their relevance for the model comparison.

Table 2
Points for output of variables.

Material	Point name	Point position, r [m]
Heater	H1	0.1575
	H2	0.42
Granular bentonite material (GBM)	GBM1	0.590625
	GBM2	0.9625
	GBM3	1.35625
Shotcrete (SC)	SC1	1.43
	SC2	1.5
	SC3	1.58
Excavation damage zone (EDZ)	EDZ1	1.63
	EDZ2	1.7
	EDZ3	1.78
Opalinus clay(OPA)	OPA1	1.83
	OPA2	2
	OPA3	5
	OPA4	20
	OPA5	30
	OPA6	40
	OPA7	50

3.2. Implementation and numerical settings

Both non-isothermal THM initial boundary value problems, i.e., TH²M and TRM, are solved using the standard Galerkin finite element method for a monolithically coupled set of equations using a Backward Euler time discretization. The mesh is subdivided into 618 quad cells along the x -axis with rotational symmetry around the y -axis. We modeled a time span of 380 years within an iteration-based time stepping scheme with a minimal timestep size $\Delta t = 1\text{ d}$. The heating phase started directly after a short equilibration period of 10 days, with a constant heat power until the end of the simulation. Each timestep was approximated using the implicit Euler method, solved partially by a standard Newton-Raphson scheme and a Newton-based trust-region dogleg method with Cauchy direction [55] to improve the numerical stability. We note that the numerical stability of the monolithic coupled TH²M model is also relatively sensitive to parameter variations, which

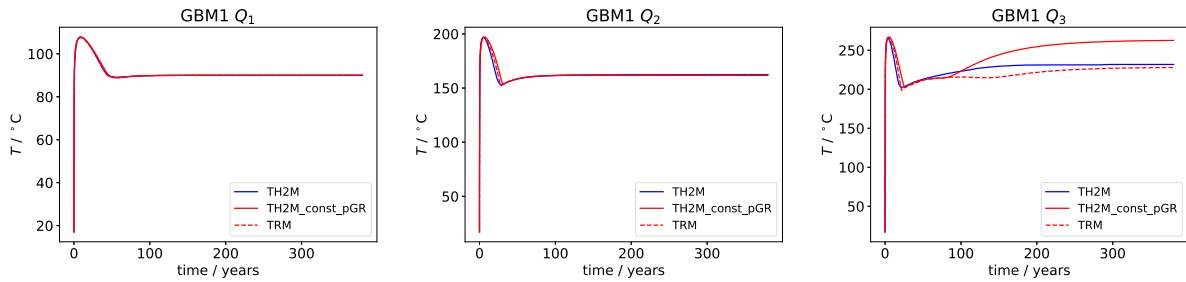


Fig. 2. Temperature curves at the closest-to-heater point for the lowest (left) and highest (right) heat power.

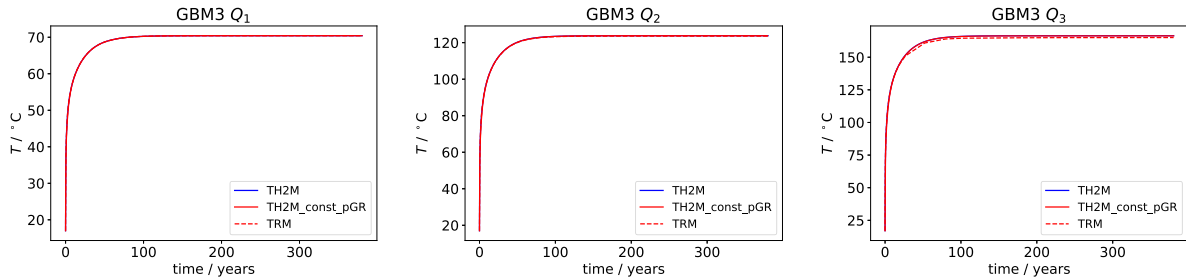


Fig. 3. Temperature curves at point GBM3 for the lowest (left) and highest (right) heat power.

we were able to improve by requiring minimal residual gas saturation for water retention behavior as well as prohibiting negative gas pressure solutions. The solver for the two-phase problem was considered converged if the overall residual norm was smaller than 7.0×10^{-5} . For the Richards based models, the relative tolerances for each component increment were set to 1×10^{-7} . The linear system of equations was solved using the Pardiso solver as implemented in the Intel® oneAPI Math Kernel Library.

4. Results and discussion

4.1. Richards-two-phase model comparison

In Fig. 2 we plotted the time–temperature curves for all models for the three tested heat powers at the closest observation point in the bentonite, as deviations between different models are expected to be highest there. We observe that the highest temperatures in all cases are reached within a few years. For the strongest heat source, temperatures at the heater surface even exceed 250°C . While safety regulators in most countries will prohibit such high temperatures in their repository concepts, we deliberately test the limits of the Richards assumption to provide confidence in the numerical tools for applications to experiments like the HotBent project, which explores the limits of bentonite buffers [26].

For the lowest heat source in Figs. 2 (left) and 3 (left), we find excellent agreement of the temperature solution between the two-phase flow model and the Richards-based model. Similarly, very good agreement is observed for the saturation solution, as seen in Figs. 4 (left) and 5 (left), and thereby for its primary variable, capillary pressure (not shown). Slight deviations appear when the saturation approaches full saturation.

For the highest heat power, we see significantly greater differences in the temperature curves (right figures in Figs. 2 and 3). This is expected, as laid out by Pitz et al. [24]. Differences in the actual implementation beyond the Richards approximation for TRM become apparent. This effect is also visible for the saturation, where the gas pressure at the heat source prevents the gas phase from vanishing. Interestingly, we observe an additional desaturation phase in all models at the closest heater point, where the models behave differently. This is also seen, to a lesser extent, in the TH²M model for the mid-range

heat power, where temperatures exceed 150°C . It is important to note that our comparison at high temperatures ($T > 150^\circ\text{C}$) only serves to cross-verify the Richards model against the two-phase flow model and does not account for effects that would lead to the breakdown of both models or require constitutive adaptations on the grounds of physical observations, such as introducing nonlinearities in the heat conduction coefficients and accounting for mineralogical changes that become relevant at high temperatures [56].

Although differences are visible between the models at different heat powers, the differences remain relatively small, especially in the low to mid-saturation regime $S_L < 0.8$ and where temperatures do not exceed 150°C .

The trends observed over time are confirmed by the behavior along the x -axis at times 15a (left) and 150a (right) in Figs. 6 and 7. For temperature (Fig. 6), no deviations are found for the lowest heat power. For the medium heat power, slight differences are visible at 15 years between the TH²M model and the other models. This behavior is reflected in the time-dependent plots as well, where temperature shows those deviations predominantly for GBM1 at early times. A similar behavior is found at the highest heat power, where differences do not entirely vanish at 150 years. Additionally, small changes are observed between the constant gas TH²M model and the TRM model, traced back to differences in the implementation of the enthalpy balance as stated in the previous paragraph.

For completeness, the saturation profiles along the x -axis are plotted in Fig. 7. Here, it becomes apparent that the lower temperatures observed for the TH²M model correspond to a faster resaturation behavior, as the liquid water increases thermal conductivity and thereby energy dissipation. These trends are explained by different expressions arising from the vapor diffusion terms in all three models, which become relevant at higher temperatures. Compare Eqs. (5) and (8) for details. The jumps observed in the saturation profiles have their origin in the different water retention capacity of the involved materials.

4.2. Vapor diffusion model comparison

The Richards-based model in OpenGeoSys, although lacking an explicit gas phase mass balance, accounts for vapor phase effects such as vapor diffusion and latent heat. In the literature, two main expressions are suggested (see Section 2.4.5 for details). Whereas the DeVries model

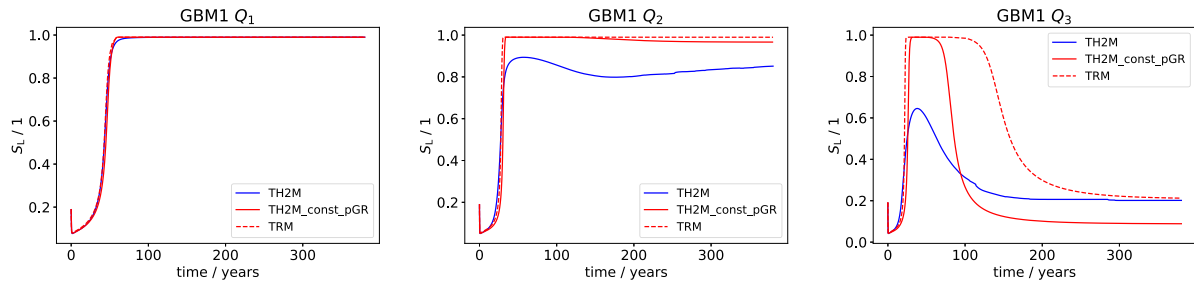


Fig. 4. Saturation curves at the closest-to-heater point for the lowest (left) and highest (right) heat power.

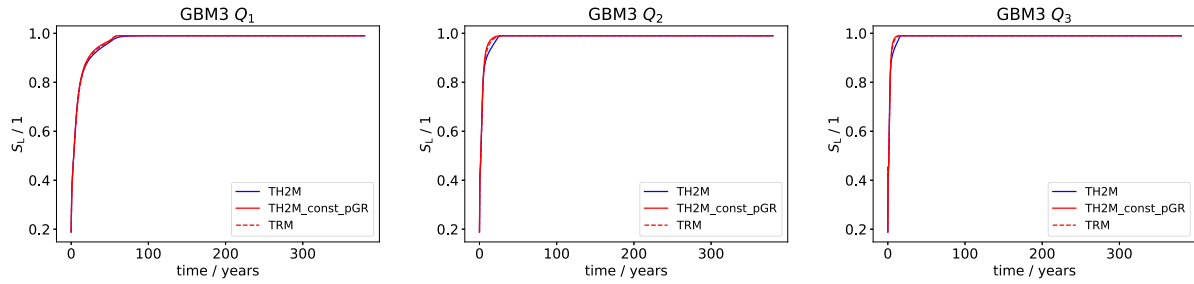


Fig. 5. Saturation curves at point GBM3 for the lowest (left) and highest (right) heat power.

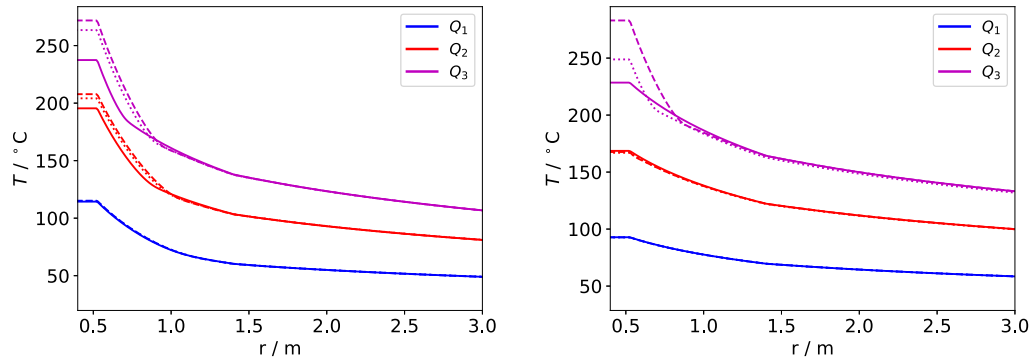


Fig. 6. Temperature profiles for TH²M (continuous line), TH²M with constant gas pressure (dashed line), and TRM model (dotted line) at a time of 15 years (left) and 150 years (right).

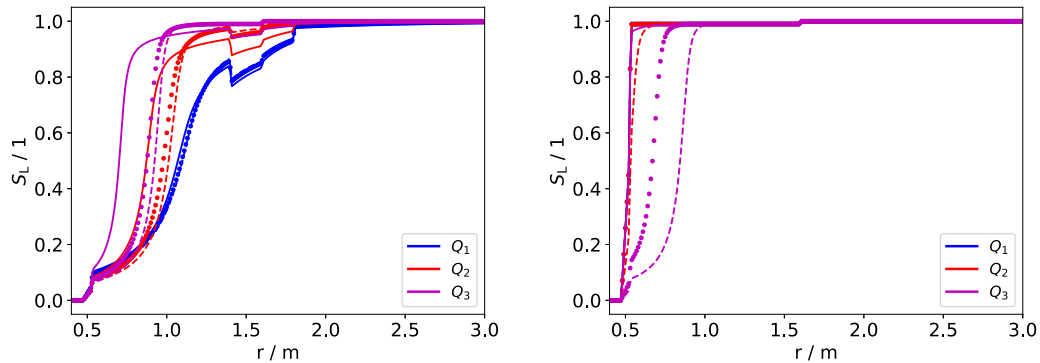


Fig. 7. Saturation profiles for TH²M (continuous line), TH²M with constant gas pressure (dashed line), and TRM model (dotted line) at a time of 15 years (left) and 150 years (right).

depends on liquid pressure and temperature, the FEBEX model, as it is called in OpenGeoSys, depends on temperature only. This makes the comparison between both models particularly interesting for our study. As both describe diffusion processes, we expect them to agree once equilibrium is reached. However, we are also interested in how both models affect de- and resaturation before equilibrium.

In Fig. 8 and Fig. 9, we plotted the temperatures at times of 15 years and 150 years, respectively. For all models, there is no significant difference between the vapor diffusion models for the lowest heat power. However, a slight difference appears for Q_2 and more pronounced differences for the highest heat power Q_3 . Here, the DeVries model leads to lower temperatures with the TH²M model but higher

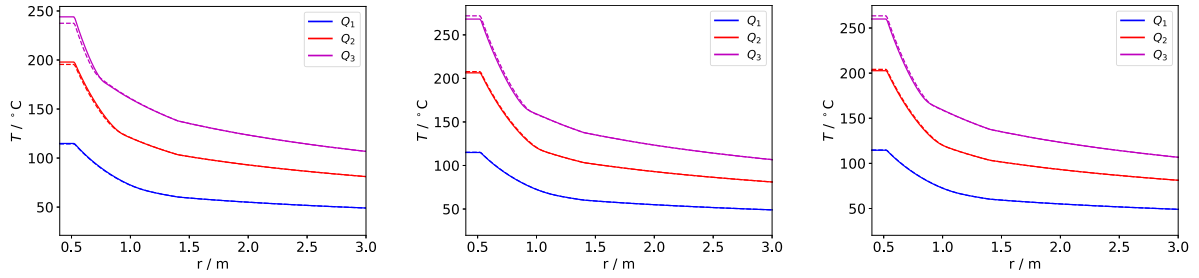


Fig. 8. Impact of the VD model on temperature at 15 years. The DeVries model (dashed line) results in lower temperatures for the TH²M model (left) but higher temperatures for the TH²M model with constant gas pressure (middle) and the TRM model (right) compared to the FEBEX model (continuous line).

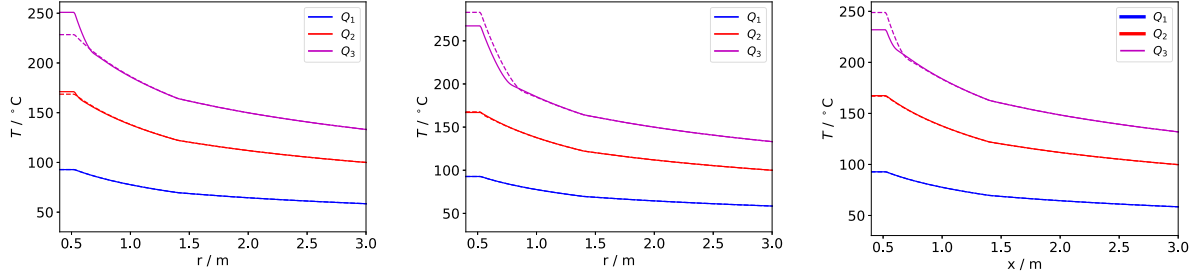


Fig. 9. Impact of the VD model on temperature at 150 years. The DeVries model (dashed line) results in lower temperatures for the TH²M model (left) but higher temperatures for the TH²M model with constant gas pressure (middle) and the TRM model (right) compared to the FEBEX model (continuous line).

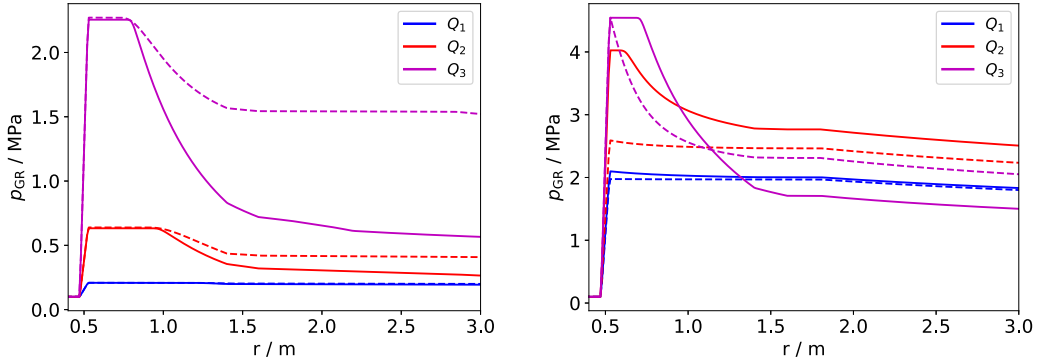


Fig. 10. Gas pressure for the TH²M model at 15 years (left) and 150 years (right). The continuous line denotes the FEBEX model, whereas the DeVries model is indicated by a dashed line.

temperatures with the constant gas pressure models.

At first sight, this seems counterintuitive. Higher gas pressure values in the TH²M model (see Fig. 10) should slow the diffusion of the vapor and thereby maintain lower saturation values, preventing heat dissipation. However, this is apparently not the case, as can be seen in the saturation plots for all three models (Figs. 11–13). Examining the temporal evolution and resaturation near the fully saturated clay interface confirms that the DeVries model leads to lower diffusivities in the TH²M model. Lower diffusivities not only slow down resaturation, but also reduce the effect of thermal diffusion, which explains the nearly full saturation for all three heat powers at 150 years (Fig. 11, right).

5. Sensitivity analysis

To underline the significance of our results, we performed a sensitivity screening for the most relevant parameters for vapor transport. For this evaluation, we used a heat power Q_2 , which we consider to produce results still within the margins, a time span of 30 years, in which most changes occur, and the De Vries vapor diffusion model.

As parameters, we included permeabilities for GBM and shotcrete, heat conductivities for GBM and shotcrete, shotcrete and GBM porosity and the base diffusion coefficient of the water vapor. The permeabilities and the diffusion coefficient were varied by a factor of 10 while the other parameters were varied by $\pm 10\%$.

We evaluated temperature and pressure results over the equally sampled time span using the measure

$$\Delta = \sqrt{\frac{1}{n} \sum_{i=1}^n (d_i^{\text{TRM}} - d_i^{\text{TH2M}})^2}. \quad (28)$$

Where n is the number of time steps and d_i indicates the variable data point for a given model at an observation point for the time step i . Although most parameters do not affect our conclusion much, we see that a high diffusion coefficient and low bentonite permeability lead to a greater (about double) model split, whereas inverse changes improve model agreement (Fig. 14). Both parameters have a slight effect on thermal diffusion but significantly affect the regime in which thermal diffusion and resaturation compete as nearly full saturation is reached. This coincides with the mismatch that is visible in Fig. 2

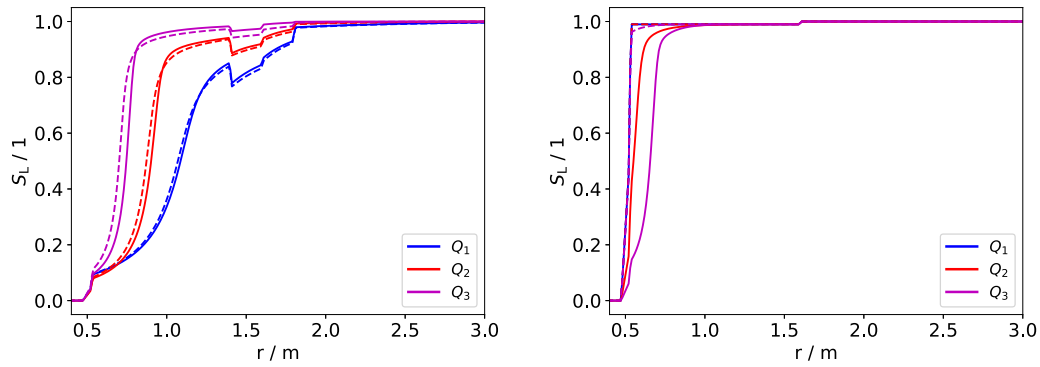


Fig. 11. Saturation for the TH²M model at 15 (left) and 150 years (right). The continuous line denotes the FEBEX model, whereas the DeVries model is indicated by a dashed line.

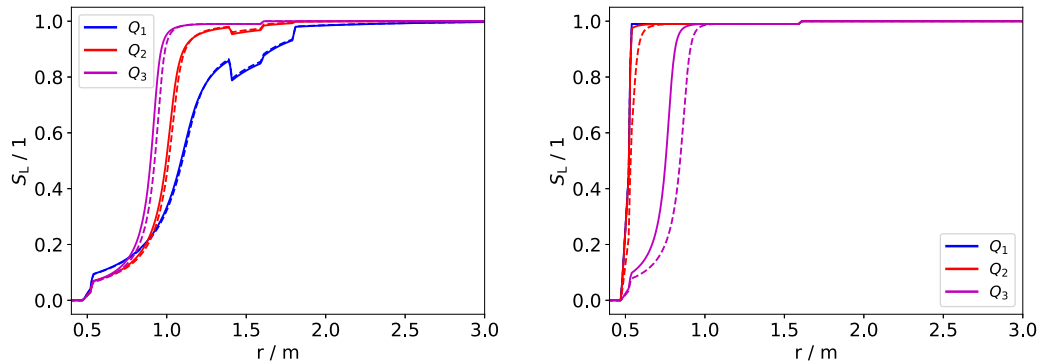


Fig. 12. Saturation for the TH²M model with constant gas pressure at 15 years (left) and 150 years (right). The continuous line denotes the FEBEX model, whereas the DeVries model is indicated by a dashed line.

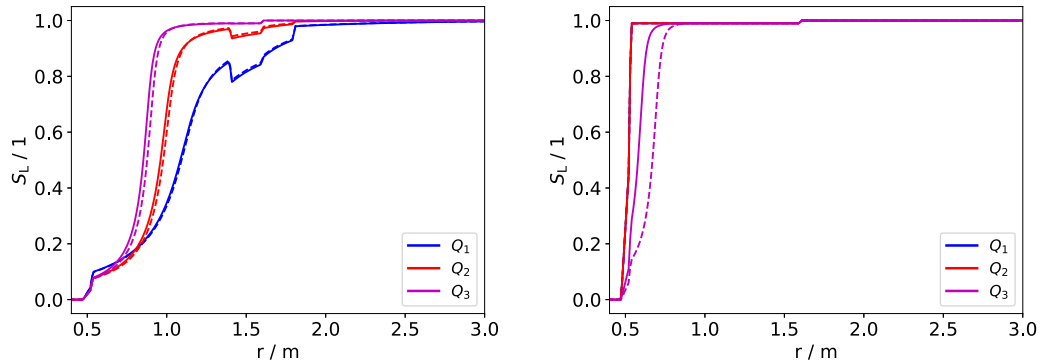


Fig. 13. Saturation for the TRM model at 15 years (left) and 150 years (right). The continuous line denotes the FEBEX model, whereas the DeVries model is indicated by a dashed line.

(middle) and 4 (middle). As our analysis indicated, the pressure and saturation responses are somewhat proportional to the temperature changes; we omit to display their results here and simply note that they are underlining our findings for the temperature response.

6. Conclusion

In our study, we demonstrated that the significantly simplified thermo-Richards-mechanics model accurately captures the behavior of the fully coupled two-phase-two-component-flow-mechanics model throughout a wide range of conditions. Although the agreement is especially good for maximum temperatures well below 200 °C, we find good agreement even for higher temperatures if the area of interest is at a significant distance from the heater, where temperatures have

significantly fallen below 200 °C, but what we still consider as near field. However, we emphasize that, under these conditions, the impact of the vapor diffusion model becomes more relevant and significantly affects the resaturation behavior. Another important factor is that gas permeabilities should be high enough to satisfy the Richards assumption, which might be an issue in some rock types where the saturation dependence of the relative permeability could lead to a local violation of the Richards assumption.

The sensitivity analysis showed that the findings of our study are valid beyond the set of parameters chosen. However, special care should be taken if the diffusion coefficients/relative liquid permeabilities in the unsaturated area of interest are significantly above/below the ones used in this study.

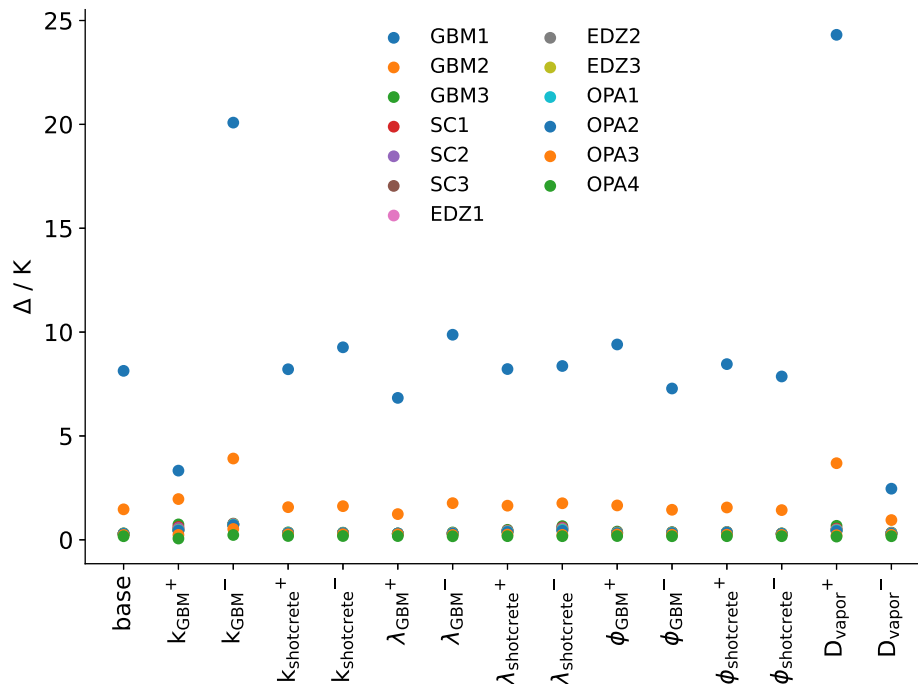


Fig. 14. Sensitivity analysis of the difference between the temperature response the TH2M and TRM model.

The results presented in this study build confidence in the use of Richard's based models that regard vapor diffusion effects in the context of thermo-hydro-mechanical modeling of radioactive waste repositories. The very good agreement between the two-phase approach and the Richards model over relatively wide parameter ranges makes the TRM model a valuable computationally less expensive alternative to fully coupled TH2M models. This is true in particular if the quantities of interest are maximum temperatures at the heater, as used in thermal dimensioning studies for repositories, as they are reached before full resaturation of the bentonite (which is where both models tend to diverge most) or if one is interested in the temperature and pressure evolution at a certain distance from the heater as differences tend to shrink with increasing distance. Furthermore, our study shows under which circumstances two-phase flow-based models should be preferred, as they lead to a more accurate prediction of the temporal evolution of the resaturation behavior of the bentonite at very high temperatures. As a last point, we would like to emphasize that for temperatures well above 150 °C, other high-temperature effects, such as mineral transformation, might be taken into account, which were not within the scope of this manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix. Overview of symbols

See Table A.3.

Data availability

Data will be made available on request.

Table A.3

Explanation of symbols. L (length), T (time), Θ (temperature), N (amount of substance) and M (mass) can represent any consistent set of units.

Symbol	Description	Dimension	Unit in use
$\circ_\alpha$	property of phase $\alpha \in \{G, L, S\}$	—	—
$\circ_\zeta$	property of component $\zeta \in \{W, C\}$	—	—
$\frac{d\circ}{dt}$	material time derivative of the solid	T^{-1}	s^{-1}
$\beta_{p,SR}$	solid compressibility	$LM^{-1}T^2$	Pa^{-1}
$\beta_{p,LR}$	liquid compressibility	$LM^{-1}T^2$	Pa^{-1}
$\beta_{c,LR}$	liquid expansivity due to gas dissolution	L^3N^{-1}	$m^3 \text{ mol}^{-1}$
α_B	Biot coefficient	—	—
$\alpha_{T,SR}$	solid thermal expansion tensor	Θ^{-1}	K^{-1}
A_ζ^c	accumulative advective mass flux of ζ	$MT^{-1}L^{-2}$	$kg \text{ s}^{-1} m^{-2}$
A_α^c	advective mass flux of ζ in α	$MT^{-1}L^{-2}$	$kg \text{ s}^{-1} m^{-2}$
A_α	advective mass flux of phase α	$MT^{-1}L^{-2}$	$kg \text{ s}^{-1} m^{-2}$
C	solid stiffness matrix	$ML^{-1}T^{-2}$	Pa^{-1}
c_L^C	concentration of dissolved gas	NL^{-3}	$mol \text{ m}^{-3}$
c_{pa}	specific heat capacity of phase α	$L^2T^{-2}\Theta^{-1}$	$J K^{-1} kg^{-1}$
$c_{p,vap}$	specific heat capacity of water vapor	$L^2T^{-2}\Theta^{-1}$	$J K^{-1} kg^{-1}$
D_ζ^c	water vapor diffusion coefficient	L^2T^{-1}	$m^2 \text{ s}^{-1}$
d_α^c	diffusion velocity of ζ in α	LT^{-1}	$m \text{ s}^{-1}$
g	gravitational acceleration acting on α	LT^{-2}	$m \text{ s}^{-2}$
H^C	Henry coefficient	$NT^2\Theta^{-2}$	$mol \text{ m}^{-3} Pa^{-1}$
h_ζ, h_α	specific enthalpy of component ζ or phase α	L^2T^{-2}	$J kg^{-1}$
J_ζ^c	accumulative diffusive mass flux of ζ	$MT^{-1}L^{-2}$	$kg \text{ s}^{-1} m^{-2}$
J_α^c	diffusive mass flux of ζ in α	$MT^{-1}L^{-2}$	$kg \text{ s}^{-1} m^{-2}$
k_{rel}^ζ	relative permeability of α	—	—
k_S	intrinsic permeability	L^2	m^2
L_0	water latent heat	ML^2T^{-2}	$J m^{-3}$
λ_{eff}	bulk thermal conductivity	$MLT^{-3}\Theta^{-1}$	$W m^{-1} K$
M_ζ^c	molar mass of ζ	MN^{-1}	$kg \text{ mol}^{-1}$
M_G	molar mass of gas mixture	MN^{-1}	$kg \text{ mol}^{-1}$
μ_{aR}	phase viscosity	$ML^{-1}T^{-1}$	$Pa \text{ s}$
ϕ	medium porosity	—	—
p_{cap}	capillary pressure	$ML^{-1}T^{-2}$	Pa
p_{GR}^C	non-aqueous component partial pressure	$ML^{-1}T^{-2}$	Pa
p_{GR}, p_{LR}	gaseous and liquid phase pressure	$ML^{-1}T^{-2}$	Pa
p_{SR}	solid pressure	$ML^{-1}T^{-2}$	Pa
p_{GR}^W	vapor pressure	$ML^{-1}T^{-2}$	Pa
p_{vap}^W	saturated vapor pressure	$ML^{-1}T^{-2}$	Pa
ρ_{aR}	phase density	ML^{-3}	$kg \text{ m}^{-3}$
ρ_{FR}^ζ	effective component density	ML^{-3}	$kg \text{ m}^{-3}$
$\rho_{GR}^\zeta, \rho_{LR}^\zeta$	density of ζ in gaseous and liquid phase	ML^{-3}	$kg \text{ m}^{-3}$
ρ_{GR}, ρ_{LR}	gaseous and liquid phase density	ML^{-3}	$kg \text{ m}^{-3}$
$\rho_{LR,ref}$	reference liquid phase density	ML^{-3}	$kg \text{ m}^{-3}$
ρ_{GR}^W	vapor density	ML^{-3}	$kg \text{ m}^{-3}$
R	universal gas constant	$ML^2T^{-2}N^{-1}\Theta^{-1}$	$J K^{-1} mol^{-1}$
S_L	liquid saturation	—	—
$S_{L,max}, S_{L,res}$	maximum and residual liquid saturation	—	—
$S_{e'}, S_{entry}$	effective liq. sat. and saturation at gas entry	—	—
σ'	effective stress tensor	$ML^{-1}T^{-2}$	Pa
T	temperature	Θ	K
T_{ref}	reference temperature	Θ	K
θ_{vap}	volumetric vapor content	—	—
u_α	specific internal energy of phase α	L^2T^{-2}	$J kg^{-1}$
u_S	solid displacement vector	L	m
$\tilde{u}_{aS}$	Darcy velocity w.r.t solid	LT^{-1}	$m \text{ s}^{-1}$
$x_{m,\alpha}^\zeta$	mass fraction of component ζ in phase α	—	—
$x_{n,\alpha}^\zeta$	molar fraction of component ζ in phase α	—	—
χ	Bishop's coefficient	—	—

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